GenAI Engineer Roadmap

An 8-week study plan with practical project milestones

Beginner-friendly study guide • Concepts, examples, and practical exercises

How to study this library

Use this order: Python fundamentals → GenAI fundamentals → Hugging Face and LLMs → RAG → application APIs and deployment → advanced evaluation and agents.

Study one concept, run the example, change one value, observe the result, and write a short explanation in your own words. Do not move on just because code ran once.

Suggested 8-week plan

Weeks 1–2: Python

Practice variables, collections, conditions, loops, functions, files, exceptions, modules, virtual environments and JSON. Build a text-file analyzer.

Week 3: GenAI foundations

Learn tokens, transformers, inference, prompting, context windows, embeddings and hallucinations. Explain each concept without notes.

Week 4: Model tooling

Understand the Hugging Face tokenizer/model pipeline, CPU constraints, generation parameters and model licenses. Keep the existing Qwen chatbot working.

Weeks 5–6: RAG

Add TXT ingestion first, then PDF and DOCX. Implement chunking, embeddings, local vector search, grounded prompting and citations.

Week 7: Engineering quality

Add error handling, logs, test questions, prompt-injection tests, source validation and a clean README.

Week 8: Share and explain

Document architecture, setup, limitations and screenshots. Add Git commits and prepare to explain the design decisions in an interview.

Portfolio project milestones

• Milestone 1: existing Qwen chatbot runs reproducibly.

• Milestone 2: one TXT file is extracted and split into inspectable chunks.

• Milestone 3: query retrieval returns useful passages.

• Milestone 4: Qwen answers from retrieved context and admits missing evidence.

• Milestone 5: PDF and DOCX support plus source citations.

• Milestone 6: tests, documentation, safe configuration and GitHub repository.

Free learning resources

Use the official Python tutorial for language details and the Hugging Face course for Transformers and LLM concepts. OpenAI's retrieval and embedding guides are useful for understanding hosted retrieval and the underlying ideas, even if your local implementation uses a different vector store.

Documentation pages are living resources and may change. This guide links to official resources rather than promising that every provider feature is free. Hosted APIs and storage can have costs; a local Qwen + local vector

index can be developed without paid API calls, subject to your hardware and software choices.

Official documentation and further reading

Python Tutorial — language fundamentals
<link href='https://docs.python.org/3/tutorial/' color='blue'>https://docs.python.org/3/tutorial/</link>

Hugging Face Learn — LLMs, Transformers and related tooling
<link href='https://huggingface.co/learn' color='blue'>https://huggingface.co/learn</link>

OpenAI Retrieval — retrieval concepts
<link href='https://developers.openai.com/api/docs/guides/retrieval' color='blue'>https://developers.openai.com/api/docs/guides/retrieval</link>

OpenAI embeddings — embeddings and semantic search
<link href='https://developers.openai.com/api/docs/guides/embeddings' color='blue'>https://developers.openai.com/api/docs/guides/embeddings</link>